Intermediate Macroeconomics

BSc. in Business Administration and BSc. in Finance and Accounting

Intermediate Macroeconomics – BMAK

Chapter 3: The Macroeconomy in the Long Run

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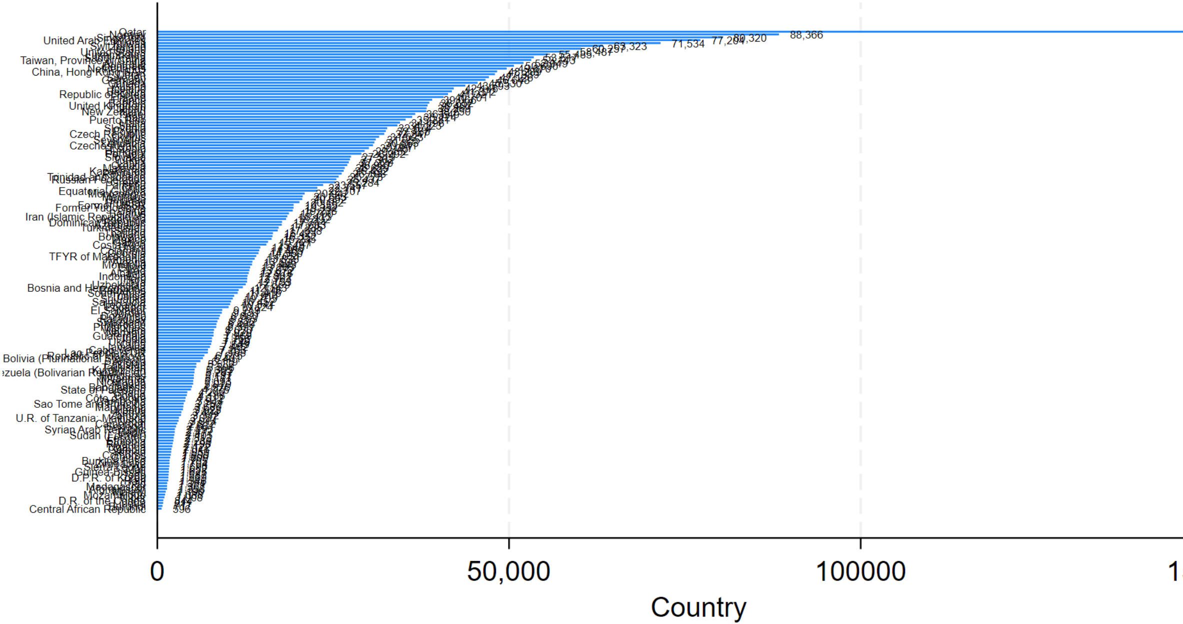
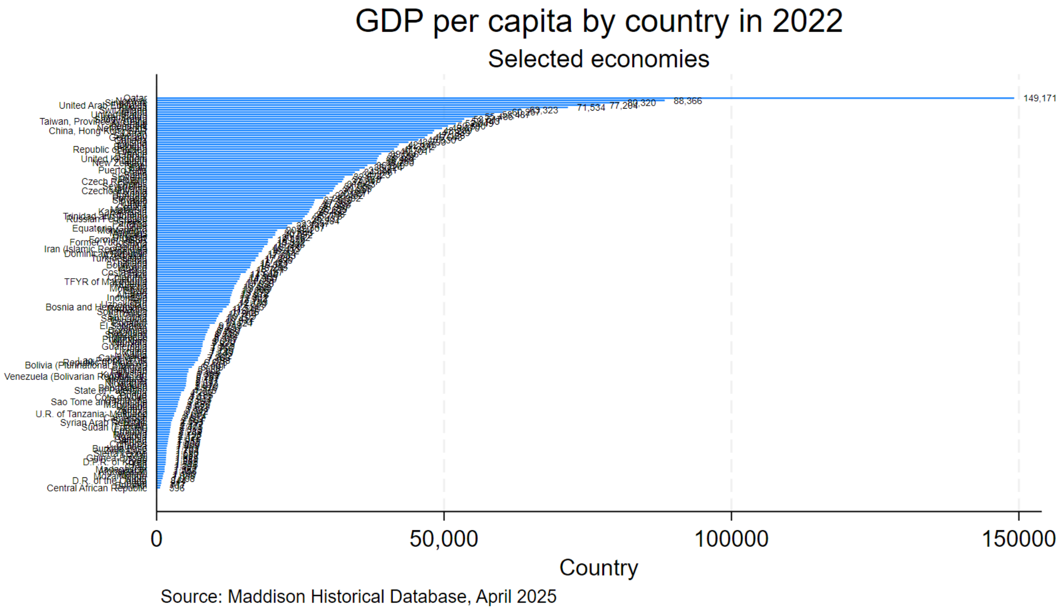
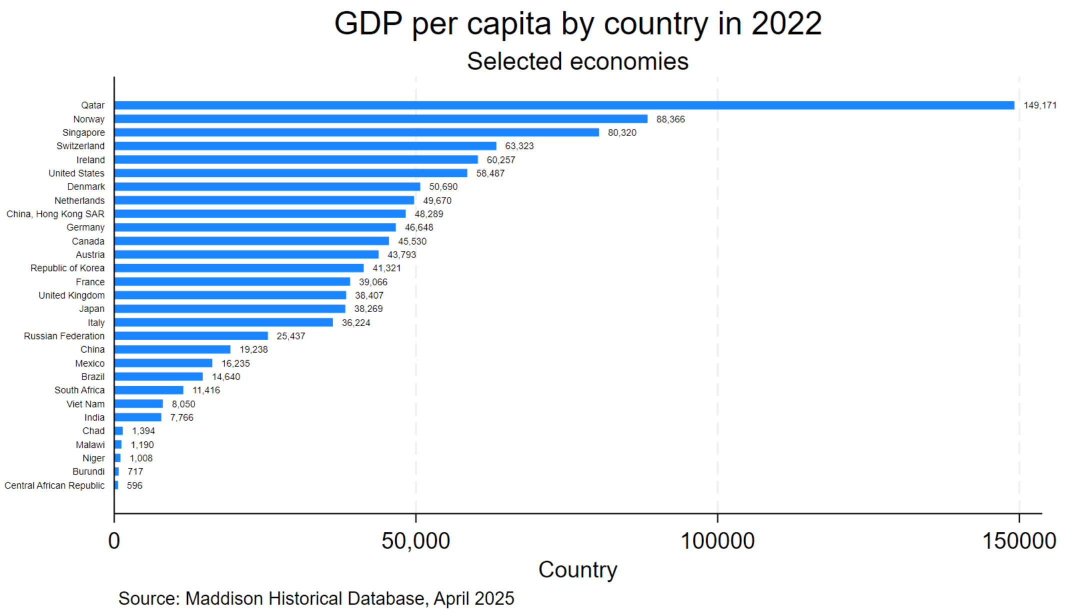


Source: Maddison Historical Database, April 2025


GDP per capita by country in 2022
Selected economies
149,171
88,366
71,534
Central African Republic
596
0
50,000
100000
150000
Country
Source: Maddison Historical Database, April 2025


GDP per capita by country in 2022
Selected economies
Qatar 149,171
Norway 88,366
Singapore 80,320
Switzerland 63,323
Ireland 60,257
United States 58,487
Denmark 50,690
Netherlands 49,670
China, Hong Kong SAR 48,289
Germany 46,648
Canada 45,530
Austria 43,793
Republic of Korea 41,321
France 39,066
United Kingdom 38,407
Japan 38,269
Italy 36,224
Russian Federation 25,437
China 19,238
Mexico 16,235
Brazil 14,640
South Africa 11,416
Viet Nam 8,050
India 7,766
Chad 1,394
Malawi 1,190
Niger 1,008
Burundi 717
Central African Republic 596
0
50,000
100000
150000
Country
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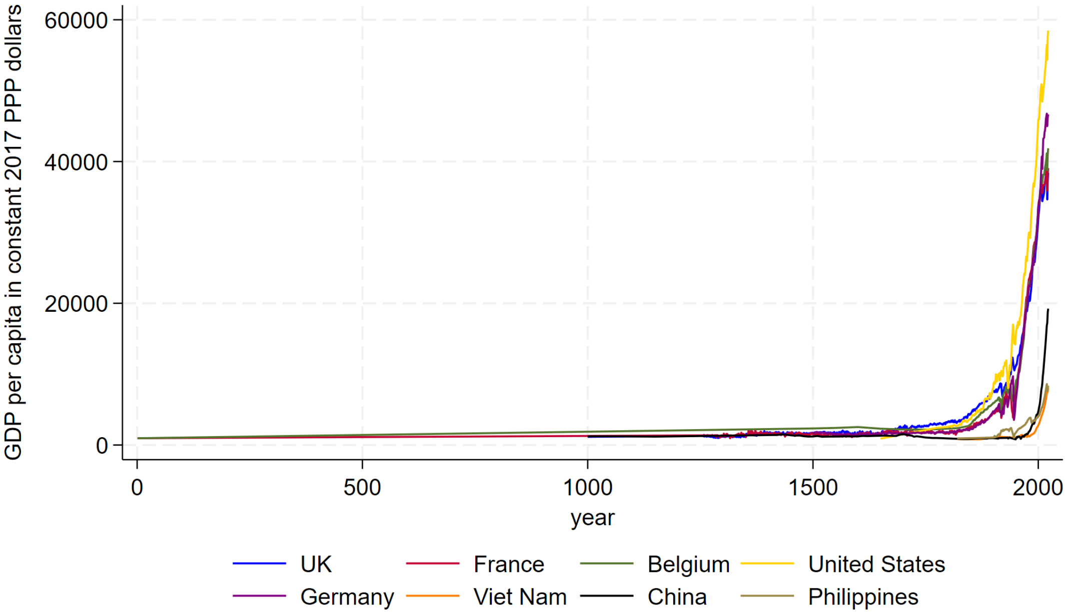
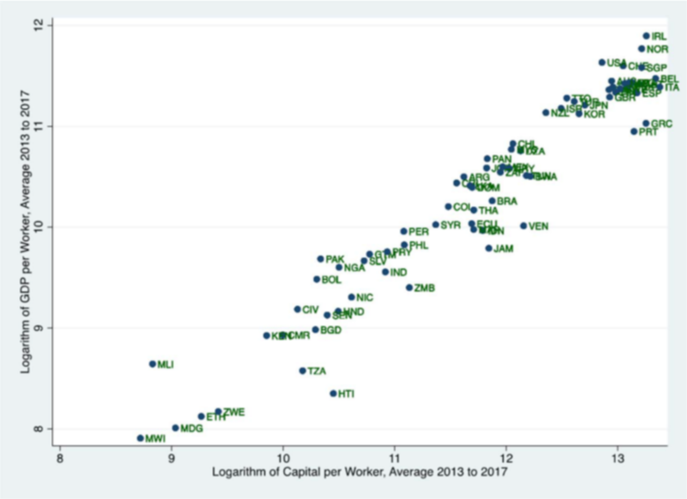

GDP per capita in constant 2017 PPP dollars
60000
40000
20000
0
0
500
1000
1500
2000
year
UK
France
Belgium
United States
Germany
Viet Nam
China
Philippines

- Sustained sizable increases in standards of living are a relatively recent phenomenon. Modern economic growth only emerged during the last two centuries.
- Sustained economic growth has emerged in different countries at different times.
- What drives the huge disparity of GDP per capita levels across countries?
  - We start with examining the data: This is why good measurement is so important!
  - Graph, analyzed, check: Kaldor's facts
    - **Stylized Fact no. 1:** Both output per capita and capital intensity keep increasing.
    - **Stylized Fact no. 2:** The capital–output ratio exhibits little or no trend over time.
    - **Stylized Fact no. 3:** Hourly wages keep rising.
    - **Stylized Fact no. 4:** The rate of profit is trendless.
    - **Stylized Fact no. 5:** The relative income shares of GDP paid to labour and capital are trendless.
  - Develop a model that is capable of replicating these facts.
  - Employ the model for further analysis.

The Solow growth model focuses on the role of physical capital – that is, equipment and structure (which includes, for example, factories, machinery, (rail-) roads, electricity networks and communication networks) – in driving cross-country differences in GDP per capita. This focus of the Solow model is empirically motivated:



An essential part of the Solow growth model is the production function for aggregate output. We thus begin with a review of such production functions:

$$\underbrace{Y_t}_{\text{output}} = F(\underbrace{A_t}_{\text{level of technology}}, \underbrace{K_t}_{\text{physical capital}}, \underbrace{L_t}_{\text{labor}}), \tag{1}$$

where for simplicity we measure L_t through the number of workers (implicitly assuming in the long run a fixed number of working hours per worker).
A special case of (1) is the Cobb-Douglas production function,

$$Y_t = K_t^{\alpha} \cdot (A_t \cdot L_t)^{1-\alpha}, \quad 0 < \alpha < 1, \tag{2}$$

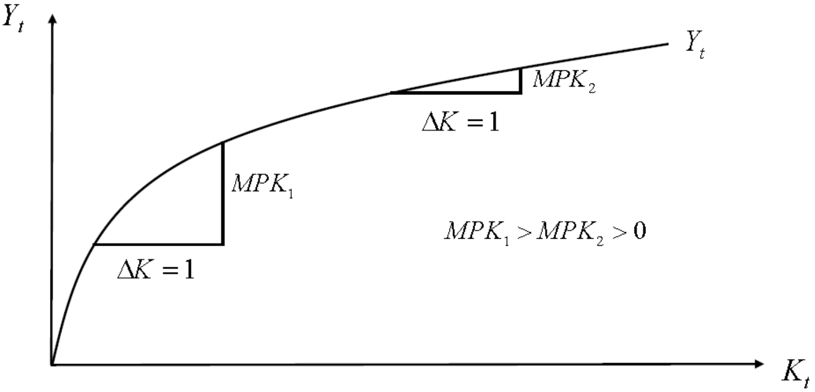
with technology affecting the level of output produced through the effectiveness of labor ("labor-augmenting technological progress"): $A_t \cdot L_t$ is often referred to as the "effective number of workers".
The marginal product of capital is defined as

$$\begin{aligned} MPK_t = \frac{\partial F(A_t, K_t, L_t)}{\partial K_t} = \frac{\partial Y_t}{\partial K_t} \\ = \frac{\partial \left[K_t^{\alpha} \cdot (A_t \cdot L_t)^{1-\alpha}\right]}{\partial K_t} \\ = \alpha \cdot K_t^{\alpha-1} \cdot (A_t \cdot L_t)^{1-\alpha} \\ = \alpha \cdot \left(\frac{K_t}{A_t \cdot L_t}\right)^{\alpha-1}, \end{aligned} \tag{3}$$

measuring the extra amount of output that is produced when one (infinitesimal) unit of capital is added, holding the other inputs constant.

Note that MPK_t is diminishing in K_t:

$$\frac{\partial MPK_t}{\partial K_t} = \frac{\partial}{\partial K_t}\left[\alpha \cdot \left(\frac{K_t}{A_t \cdot L_t}\right)^{\alpha-1}\right] = \alpha \cdot (\alpha - 1) \cdot \left(\frac{K_t}{A_t \cdot L_t}\right)^{\alpha-2} \cdot \frac{1}{A_t \cdot L_t} < 0. \tag{4}$$



The capital accumulation equation (introduced when discussing the macroeconomic accounts):

$$K_{t+1} = K_t - \underbrace{\delta}_{\substack{\text{rate of depreciation} \\ \text{of physical capital}}} \cdot K_t + \underbrace{I_t}_{\text{gross investment}},$$

$$= (1-\delta) \cdot K_t + I_t, \qquad 0 < \delta < 1, \tag{5}$$

and a link between firm and household behavior:

$$S_t = S_t^{pvt} = I_t. \tag{6}$$

From our discussion of the macroeconomic accounts, we know that (6) is satisfied if $S_t^{govt} = 0$ and $CA_t = 0$.
The simple Solow growth model also specifies that

$$S_t = \underbrace{s}_{\text{rate of saving}} \cdot Y_t, \qquad 0 < s < 1. \tag{7}$$

+ population growth (implying worker growth, that is, growth of L_t) and
+ technological progress (growth of A_t):

  Worker Growth:

$$L_{t+1} = (1 + \underbrace{n}_{\text{rate of worker growth}}) \cdot L_t. \tag{8}$$

  Technological Progress:

$$A_{t+1} = (1 + \underbrace{g}_{\text{rate of technological progress}}) \cdot A_t. \tag{9}$$

To keep analysis of this new model as straightforward as possible, we will employ a trick to **convert the new model** the capital-intensive form:

- The Cobb-Douglas production function (2),

$$Y_t = K_t^{\alpha} \cdot (A_t \cdot L_t)^{1-\alpha}, \qquad 0 < \alpha < 1,$$

  suggests that Y_t will now grow in the long run, as both A_t and L_t are growing. There will then be no steady state in Y_t.
- It seems possible, though, that there will be a **steady state in per effective worker magnitudes**, namely, magnitudes obtained by dividing level magnitudes (such as output Y_t) by the product of A_t and L_t:

$$\underbrace{\frac{Y_t}{A_t \cdot L_t}}_{\text{output per effective worker}} = \frac{K_t^{\alpha} \cdot (A_t \cdot L_t)^{1-\alpha}}{A_t \cdot L_t} = \left(\frac{K_t}{A_t \cdot L_t}\right)^{\alpha}. \tag{10}$$

- Rather than directly analyzing the dynamics of K_t (as in the simple Solow growth model), we will thus now first analyze the dynamics of

$$\underbrace{\tilde{K}_t}_{\text{capital per effective worker}} = \frac{K_t}{A_t \cdot L_t}. \tag{11}$$

+ Once we have obtained the steady state for $\tilde{K}_t$, it will be straightforward to determine the dynamics of empirically interesting magnitudes such as Y_t, as from Equations (10) and (11) we have that

$$Y_t = A_t \cdot L_t \cdot \underbrace{\frac{Y_t}{A_t \cdot L_t}}_{=\tilde{Y}_t} = A_t \cdot L_t \cdot (\tilde{K}_t)^{\alpha}. \tag{12}$$

The key equation to understand the simple Solow model was Equation (9), which can be written as

$$\begin{aligned} \Delta K_{t+1} &= -\delta \cdot K_t + s \cdot Y_t \\ \Leftrightarrow \quad K_{t+1} &= (1-\delta) \cdot K_t + s \cdot Y_t \\ \Leftrightarrow \quad K_{t+1} &= (1-\delta) \cdot K_t + s \cdot K_t^{\alpha} \cdot (A_t \cdot L_t)^{1-\alpha}. \end{aligned} \tag{13}$$

To transform (13) to per effective worker magnitudes, we divide on both sides of this equation by $A_t \cdot L_t$:

$$\underbrace{\frac{K_{t+1}}{A_t \cdot L_t}}_{\substack{= \frac{K_{t+1}}{A_{t+1} \cdot L_{t+1}} \cdot \frac{A_{t+1}}{A_t} \cdot \frac{L_{t+1}}{L_t} \\ = \tilde{K}_{t+1} \cdot (1+g) \cdot (1+n) \\ \approx (1+g+n) \cdot \tilde{K}_{t+1}}} = (1-\delta) \cdot \underbrace{\frac{K_t}{A_t \cdot L_t}}_{=\tilde{K}_t} + \underbrace{\frac{s \cdot K_t^{\alpha}}{A_t \cdot L_t} \cdot (A_t \cdot L_t)^{1-\alpha}}_{=s \cdot (\tilde{K}_t)^{\alpha}}. \tag{14}$$

Re-arranging (14), we have

$$\tilde{K}_{t+1} = \left(\frac{1-\delta}{1+g+n}\right) \cdot \tilde{K}_t + \left(\frac{s}{1+g+n}\right) \cdot (\tilde{K}_t)^{\alpha}, \tag{15}$$

or

$$\begin{aligned}\Delta\tilde{K}_{t+1} = \tilde{K}_{t+1} - \tilde{K}_t &= \left(\frac{s}{1+g+n}\right) \cdot (\tilde{K}_t)^{\alpha} + \left(\frac{1-\delta}{1+g+n} - 1\right) \cdot \tilde{K}_t, \\ &= \left(\frac{1}{1+g+n}\right) \cdot \left[s \cdot (\tilde{K}_t)^{\alpha} - (\delta+g+n) \cdot \tilde{K}_t\right]. \end{aligned} \tag{16}$$

Thus, capital per effective worker is in steady state, that is,

$$\Delta\tilde{K}_{t+1} = \tilde{K}_{t+1} - \tilde{K}_t = 0, \tag{17}$$

if

$$s \cdot (\tilde{K}_t)^{\alpha} - (\delta+g+n) \cdot \tilde{K}_t = 0. \tag{18}$$

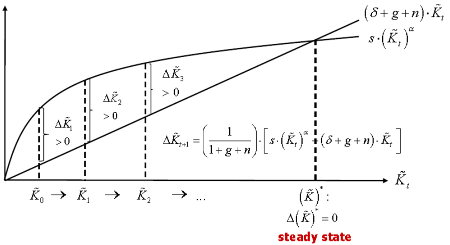
We can now analyze the Solow growth model with population growth and technological progress:
We plot separately the schedules of

+ gross investment per effective worker,

$$s \cdot (\tilde{K}_t)^{\alpha}, \tag{19}$$

+ the amount of investment per effective worker that is needed to keep capital per effective worker constant, to account, in particular, for losses of capital per effective worker due to depreciation, population growth and technological progress:

$$(\delta + g + n) \cdot \tilde{K}_t. \tag{20}$$



Note that for any initial level of capital per effective worker $\tilde{K}_0$ less than $(\tilde{K})^*$, the capital stock per effective worker always converges to $(\tilde{K})^*$. Furthermore, at $(\tilde{K})^*$ the amount of capital per effective worker remains constant, as net investment in capital per effective worker is equal to zero.

Note that for any initial level of capital per effective worker $\tilde{K}_0$ larger than $(\tilde{K})^*$, the capital stock per effective worker always converges to $(\tilde{K})^*$. Furthermore, at $(\tilde{K})^*$ the amount of capital per effective worker remains constant, as net investment in capital per effective worker is equal to zero.

We have thus established that for any Solow model economy with a strictly positive initial level of capital per effective worker, the unique long-run outcome for the capital stock per effective worker is the steady-state level $(\tilde{K})^*$.
From Equation (20), we may obtain this steady-state capital stock analytically:

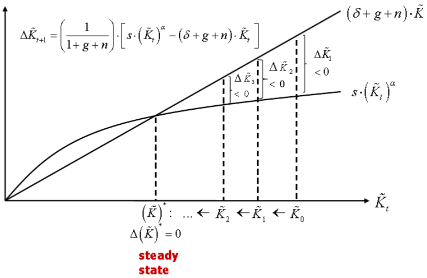
$$\Delta\tilde{K}_{t+1} = 0 \quad \Leftrightarrow \quad s \cdot (\tilde{K}_t)^\alpha = (\delta + g + n) \cdot \tilde{K}_t \quad \Leftrightarrow \quad \frac{s}{\delta + g + n} = [(\tilde{K})^*]^{1-\alpha}$$

$$\Leftrightarrow \quad (\tilde{K})^* = \left(\frac{s}{\delta + g + n}\right)^{\frac{1}{1-\alpha}} . \tag{21}$$

Substituting into (18), output per effective worker in the long run is also constant and is given by:

$$(\tilde{Y})^* = \left((\tilde{K})^*\right)^\alpha = \left(\frac{s}{\delta + g + n}\right)^{\frac{\alpha}{1-\alpha}} . \tag{22}$$

Another look at the **steady state for the Solow growth model with population growth and technological progress**:



According to the Solow growth model with population growth and technological progress, what is the **long-run growth rate of output per effective worker**, $Y_t/(A_t \cdot L_t)$?

In the long run, capital per effective worker, $\tilde{K}_t$, reaches steady state. Also remember from (18) that output per effective worker is given by

$$\frac{Y_t}{A_t \cdot L_t} = (\tilde{K}_t)^\alpha.$$

Therefore we have that

$$\textit{growth rate}\left(\frac{Y_t}{A_t \cdot L_t}\right) = \textit{growth rate}\left[(\tilde{K}_t)^\alpha\right] = 0. \tag{23}$$

As in steady state $\tilde{K}_t = \tilde{K}_{t-1}$, the long-run growth rate of output per effective worker is thus equal

We thus have the following first **key prediction of the Solow growth model with population growth and technological progress**:

+ There is **no long-run growth of output per effective worker**, $Y_t/(A_t \cdot L_t)$, (rather, output per effective worker converges to a steady state).

In practice, our data and substantive economic interest are on output per worker, Y_t/L_t, and output, Y_t, though. What does the Solow growth model with population growth and technological progress predict for these variables?

+ **Output per worker**, Y_t/L_t, in the **long run grows at the rate of technological progress**, g.

To verify this, observe that

$$\frac{Y_t}{L_t} = A_t \cdot \underbrace{\frac{Y_t}{A_t \cdot L_t}}_{=\tilde{Y}_t}, \tag{24}$$

in turn implying that

$$\begin{aligned} growth\ rate\left(\frac{Y_t}{L_t}\right) &= \frac{Y_t/L_t}{Y_{t-1}/L_{t-1}} - 1 = \frac{A_t \cdot \tilde{Y}_t}{A_{t-1} \cdot \tilde{Y}_{t-1}} - 1 \\ &= \frac{(1+g) \cdot A_{t-1}}{A_{t-1}} \cdot \frac{\tilde{Y}_t}{\tilde{Y}_{t-1}} - 1 = (1+g) \cdot \frac{\tilde{Y}_t}{\tilde{Y}_{t-1}} - 1 = g. \end{aligned} \tag{25}$$

As in steady state output per effective worker is constant, $\tilde{Y}_t = \tilde{Y}_{t-1}$, the long-run growth rate of output per worker is equal to g.

The resultant long-run time path of output per worker is called the **balanced growth path**. If output per effective worker had been in steady state from, say, period $t = 0$, this balanced growth path from (24) is given by

$$\left(\frac{Y_t}{L_t}\right)_{\text{balanced growth path}} = A_t \cdot (\tilde{Y})^*, \quad t = 0, 1, \ldots, \tag{26}$$

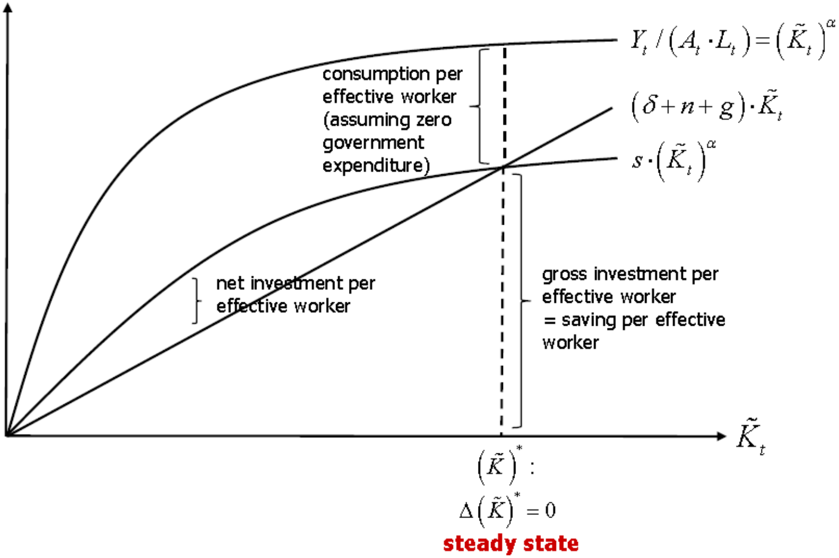
where $(\tilde{Y})^*$ denotes the steady-state value of output per effective worker. As

$$\begin{aligned} A_t &= (1+g) \cdot A_{t-1} \\ &= (1+g) \cdot [(1+g) \cdot A_{t-2}] = (1+g)^2 \cdot A_{t-2} \\ &= \cdots = (1+g)^t \cdot A_0, \end{aligned} \tag{27}$$

we have

$$\left(\frac{Y_t}{L_t}\right)_{\text{balanced growth path}} = (1+g)^t \cdot A_0 \cdot (\tilde{Y})^*. \tag{28}$$

$$\ln\left[\left(\frac{Y_t}{L_t}\right)_{\textit{balanced growth path}}\right] = t \cdot \ln(1+g) + \ln(A_0) + \ln\left[(\tilde{Y})^*\right]. \tag{29}$$



- **Total output**, Y_t, in the **long run grows** at rate $g + n$, the **sum of the rates of technological progress and the rate of worker growth**.

To verify this, observe that

$$Y_t = A_t \cdot L_t \cdot \underbrace{\frac{Y_t}{A_t \cdot L_t}}_{=\tilde{Y}_t}, \tag{30}$$

in turn implying that

$$\begin{aligned} \textit{growth rate}\,(Y_t) &= \frac{Y_t}{Y_{t-1}} - 1 = \frac{A_t \cdot L_t \cdot \tilde{Y}_t}{A_{t-1} \cdot L_{t-1} \cdot \tilde{Y}_{t-1}} - 1 \\ &= \frac{(1+g) \cdot A_{t-1}}{A_{t-1}} \cdot \frac{(1+n) \cdot L_{t-1}}{L_{t-1}} \cdot \frac{\tilde{Y}_t}{\tilde{Y}_{t-1}} - 1 = (1+g) \cdot (1+n) \cdot \frac{\tilde{Y}_t}{\tilde{Y}_{t-1}} - 1 \approx g + n. \end{aligned} \tag{31}$$

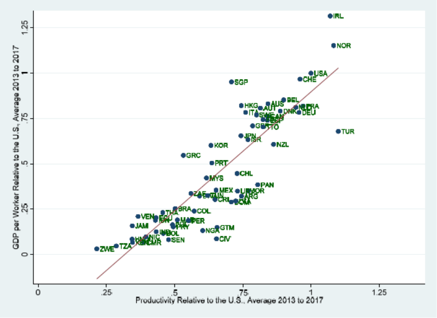
As in steady state output per effective worker is constant, $\tilde{Y}_t = \tilde{Y}_{t-1}$, the long-run growth rate of output is approximately equal to $g + n$.

+ Generally, the long-run level of output will vary across economies, and is a function of each economy's production technology (the values of α, δ, and g), its rate of saving (the value of s), its rate of worker growth (n), and its initial levels of technology (A_0) and of the number of workers (L_0).
+ **Poorer economies** (starting off at lower levels of output) **may catch up with richer ones** (starting off at higher levels of output). Put differently: Once we have filtered out cross-country differences in α, δ, g, s, n, A_0, and L_0, we will observe convergence (**conditional convergence**).

Key feature of the model driving these results: the **diminishing** (but positive) **marginal product of capital**.

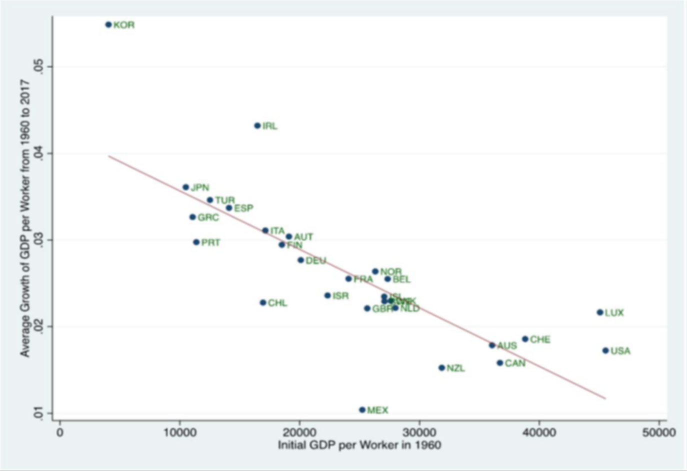
How Does the Solow Growth Model with Population Growth and Technological Progress Fare in the Data? **Role of Technological Progress**

A country's level of productivity (relative to that of the United States) is closely aligned with its level of GDP per worker (relative to that of the United States). This is consistent with the Solow growth model with population growth and technological progress.



Source of Data: Penn World Table 10.1

Consistent with the Solow growth model with population growth and technological progress, when considering a sample of countries for which differences in technology are relatively minor, countries that used to be relatively worse-off exhibit higher growth rates of GDP per worker than countries that used to be relatively better off, and there is convergence of GDP per worker.



Source of Data: Penn World Table 10.1

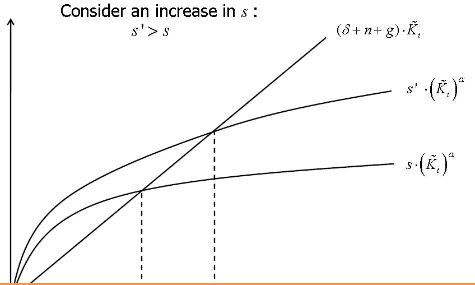
Given this favorable empirical evidence for the Solow growth model with population growth and technological progress, let us put further effort into studying the model's properties, and examine how changes in the model parameters affect its long-run balanced growth path of output per worker.

Let us **consider**, as one example, **an increase in the rate of saving**. From (36), the logarithm of the long-run balanced growth path of output per worker is given by

$$\ln\left[\left(\frac{Y_t}{L_t}\right)_{\text{balanced growth path}}\right] = t\cdot\ln(1+g)+\ln(A_0)+\ln\left[\left(\tilde{Y}\right)^*\right],$$

and thus an increase in the rate of saving may affect this growth path at most through affecting steady-state output per effective worker, $\left(\tilde{Y}\right)^*$.

Consider thus the effects of an increase in the rate of saving on the steady state in per effective worker units:



Note that the higher steady-state level of capital per effective worker,

$$\left(\tilde{K}^*\right)',$$

implies a higher steady-state level of output per effective worker also:

$$\left(\tilde{Y}^*\right)' = \left(\frac{Y}{A \cdot L}\right)'^* = \left(\left(\tilde{K}^*\right)'\right)^\alpha > \left(\tilde{K}^*\right)^\alpha = \left(\frac{Y}{A \cdot L}\right)^* = \tilde{Y}^*. \tag{32}$$

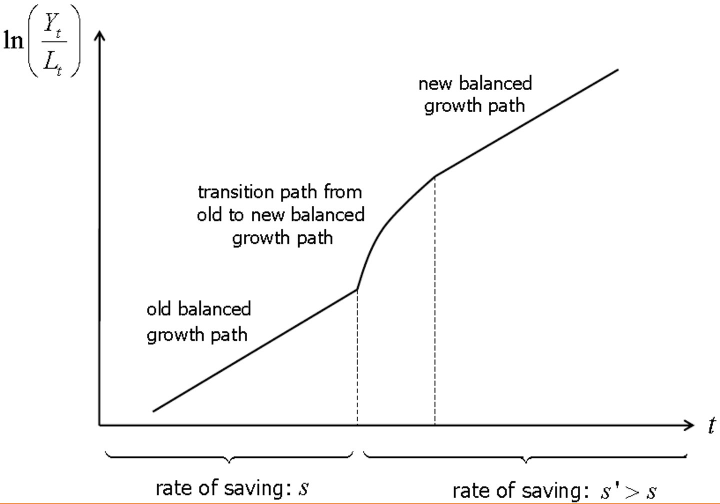
Thus, the new balanced growth path involves a higher level of output per worker:

$$\begin{aligned} \ln\left[\left(\frac{Y_t}{L_t}\right)_{\textit{new balanced growth path}}\right] &= t \cdot \ln(1+g) + \ln(A_0) + \ln\left[\left(\tilde{Y}^*\right)'\right] \\ &> t \cdot \ln(1+g) + \ln(A_0) + \ln\left[\tilde{Y}^*\right] = \ln\left[\left(\frac{Y_t}{L_t}\right)_{\textit{old balanced growth path}}\right]. \end{aligned} \tag{33}$$

Note that both the old and the new balanced growth paths in logarithms have slope (approximately) equal to g: output per worker in the long run grows at the rate of technological progress, g.

The transition from the old to the new balanced growth path is gradual, reflecting the gradual adjustment from the old steady-state level of output per effective worker to the new steady-state level of output per effective worker.

It is beyond the scope of this course to derive the curvature of the **transition path from the old to the new balanced growth path**. We confine ourselves to graphing the complete path of output per worker before and after the increase in the rate of saving:



What is the Optimal Rate of Saving in the Solow Growth Model with Population Growth and Technological Progress?

The "golden rule" aims to strike an optimal trade-off between saving and consumption by maximizing the steady-state level of consumption per effective worker.

Note that maximizing the steady-state level of consumption per effective worker is equivalent to maximizing the level of consumption along the balanced growth path, as technological progress is exogenous in the Solow growth model.

Assuming that government expenditure per effective worker is equal to zero, consumption per effective worker is equal to output per effective worker minus gross investment per effective worker. The golden rule therefore requires to maximize:

$$\max_{s}\left\{\left(\frac{C}{A\cdot L}\right)^{*}=\left(\frac{Y}{A\cdot L}\right)^{*}-\left(\frac{I}{A\cdot L}\right)^{*}\right\}. \tag{34}$$

Substituting from Equation (29) that

$$\left(\frac{Y}{A\cdot L}\right)^{*}=(\tilde{Y})^{*}=\left(\frac{s}{\delta+g+n}\right)^{\frac{\alpha}{1-\alpha}}, \tag{35}$$

and noting with the help of Equations (24) and (28) that

$$\left(\frac{I}{A\cdot L}\right)^{*}=(\tilde{I})^{*}=s\cdot\left[(\tilde{K})^{*}\right]^{\alpha}=(\delta+g+n)\cdot(\tilde{K})^{*}=(\delta+g+n)\cdot\left(\frac{s}{\delta+g+n}\right)^{\frac{1}{1-\alpha}}, \tag{36}$$

(34) can be re-written as

$$\max_{s} \left\{ \left(\frac{s}{\delta + g + n} \right)^{\frac{\alpha}{1-\alpha}} - (\delta + g + n) \cdot \left(\frac{s}{\delta + g + n} \right)^{\frac{1}{1-\alpha}} \right\}. \tag{37}$$

The first-order condition is given by:

$$\left(\frac{\alpha}{1-\alpha} \right) \cdot \left(\frac{s_{\text{golden}}}{\delta + g + n} \right)^{\frac{\alpha}{1-\alpha} - 1} \cdot \left(\frac{1}{\delta + g + n} \right) = \left(\frac{1}{1-\alpha} \right) \cdot \left(\frac{s_{\text{golden}}}{\delta + g + n} \right)^{\frac{1}{1-\alpha} - 1}. \tag{38}$$

Simplifying (44), we obtain:

$$\begin{aligned} &\left(\frac{\alpha}{1-\alpha} \right) \cdot \left(\frac{s_{\text{golden}}}{\delta + g + n} \right)^{\frac{2\alpha-1}{1-\alpha}} \cdot \left(\frac{1}{\delta + g + n} \right) - \left(\frac{1}{1-\alpha} \right) \cdot \left(\frac{s_{\text{golden}}}{\delta + g + n} \right)^{\frac{\alpha}{1-\alpha}} = 0 \\ \Leftrightarrow &\left(\frac{1}{1-\alpha} \right) \cdot \left(\frac{s_{\text{golden}}}{\delta + g + n} \right)^{\frac{\alpha}{1-\alpha}} \cdot \left[\left(\frac{\alpha}{\delta + g + n} \cdot \frac{s_{\text{golden}}}{\delta + g + n} \right)^{\frac{\alpha-1}{1-\alpha}} - 1 \right] = 0 \\ \Leftrightarrow &\quad s_{\text{golden}} = \alpha. \end{aligned} \tag{45}$$

Economic reasoning for the golden-rule saving rate: The larger α, the less the marginal product of capital diminishes as capital is accumulated, and thus the higher the return to saving. Thus the optimal rate of saving increases with α.